

Research Paper

The Certainty of Uncertainty: GIS and the Limits of Geographic Knowledge

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Abstract

Considerable effort has been devoted over the years to fighting uncertainty in geographic information in its different manifestations. Thus far, research on handling inaccuracy, fuzziness, error and related issues has focused for the most part on problems with spatial data and their direct products, typically representations of spatial objects or fields. This paper seeks to broaden the discussion of uncertainty in the geospatial domain by shifting the focus from information to knowledge. It turns out that there is a surprising number of things that we cannot know (or questions we cannot answer) that are not the result of imperfect information. Forms of not knowing are pervasive in domains as diverse as mathematics, logic, physics, and linguistics, and are apparently irreducible. This being the case it may help to explore how these realms of ignorance may affect our efforts. The paper distinguishes three different modes or forms of geospatial knowledge production, and argues that each of them has built-in imperfections, for reasons of logical principle and not just empirical fact. While much can and needs to be done to manage and resolve uncertainties where possible, I argue for accepting that uncertainty is an intrinsic property of complex knowledge and not just a flaw that needs to be excised.

1 Introduction

[There are] “two types of information-processing system: the type that is capable of converting the information it receives into knowledge and the type that is not” (Dretske 1981, p. 172)

The first kind of information-processing system, the kind capable of converting information into knowledge, is we; the GIS is of the second kind. This paper is about the symbiotic relationship between these two kinds of information systems, the cognitive

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and the digital, and about the properties of the knowledge that the former constructs with the help of the latter. While the bulk of the Geographic Information Science (GIScience) literature focuses on the strengths of that knowledge, this paper focuses on its imperfections.

Perfect knowledge is one of humanity's oldest ideals. Knowing something for certain greatly facilitates rational action. Being right about something is nearly always better than the alternative. In applications of GIS and other geo-spatial technologies, being right (accurate, correct, precise . . .) is considered of paramount importance and may indeed sometimes mean the difference between life and death. As a result considerable effort has been devoted over the years to fighting imperfect information in its different manifestations. In GIScience, research on handling uncertainty, error, inaccuracy, imprecision, ambiguity, vagueness, and related issues has focused for the most part on problems with spatial data and their direct products, typically representations of spatial objects or fields. Further, a number of researchers have pointed out that there are several other dimensions to the quality issue in the geospatial domain over and above data quality and its direct implications. Product quality has sometimes been discussed in the literature as the most critical issue and one that is only partially related to the quality of the data (Couclelis 1992). A related consideration is the improper use of geographic information as a result of lack of skills, confusion, misunderstanding, and so on, on the part of the GIS user (Hunter and Reinke 2000). Virtually all that work treats geospatial uncertainty as the result of empirical failings due to practical, technical, intellectual, psychological, social, and other such reasons.

This paper seeks to broaden the discussion of uncertainty in the geospatial domain by shifting the focus from information to knowledge. It turns out that there are a surprising number of things that we cannot know and that are not the result of human limitations. This being the case it may help to explore how these realms of ignorance may affect our efforts. **Section 2 discusses geospatial uncertainty in the context of a variety of forms of imperfect knowledge that do not appear to be a matter of incomplete, inaccurate, imprecise, or otherwise invalid information.** Such forms of not knowing are pervasive in domains as diverse as mathematics, logic, physics, linguistics, and information theory, and apparently irreducible. Drawing from work in artificial intelligence, Section 3 distinguishes three different modes or forms of geo-spatial knowledge production, and argues that each of them has built-in imperfections for reasons of logical principle and not just empirical fact. While much can and needs to be done to manage and resolve the uncertainties where possible, I argue for accepting that uncertainty is an intrinsic property of knowledge and not just a flaw that needs to be excised.

2 GIS and the Production of Geospatial Knowledge

2.1 *From Information to Knowledge (and Ignorance)*

Most definitions of GIS stress its functions of data storage, manipulation, management, visualization, and analysis, with an increasing emphasis on the latter two aspects. Less often discussed is the fact that a GIS is also a tool for the *synthesis* of geographic information, and therefore, for the production of applied geographic knowledge. GIS users can combine and manipulate information from diverse data sources, available in a variety of different formats, to generate maps, tables, graphics, and other products

that in many cases, until recently only experts could deliver (Couclelis 1992). This democratization of the knowledge production process did not come without its own problems, notable among which are the general lack of quality standards for GIS products derived by non-experts, and the risk of user misinterpretation of otherwise valid products. Unlike industrial and other products of material processes, knowledge products do not carry with them the evidence of their own inadequacy. If the new stereo does not sound right or if the cake has too much salt, you know it at once; if the model airplane you just built cannot fly, you will soon find out. But the flaws of a bad GIS knowledge product are often no more evident than those of a bad theory of navigation. Bad knowledge and erroneous beliefs do not necessarily look or behave different from good unless and until some special situation arises. Information technology differs from ordinary, material technology in that the most critical aspects of the quality of its products are often only testable through their indirect and sometimes remote consequences.

Serious though these non-expert user issues may be, this paper is about the limitations of the knowledge synthesized by experts. Thrift (1985) discusses five forms of not knowing: that which is unknown or unsuspected; that which is not understood; that which is undiscussed or undisputed; that which is deliberately hidden; and that which is distorted. To these we may add at least one more: that which one does not want to know. In varying degrees, all these forms of not knowing may affect expert as well as non-expert geospatial knowledge producers. The sampling scheme may have missed some crucial measurements; the analysis may have overlooked a critical pattern; some common methods may be taken too much for granted; some information may not be available for privacy or security reasons; the data may be of poor quality; political or ethical sensitivities may preclude the scientific investigation of a phenomenon. These are mostly questions of data quantity, quality and availability, questions of techniques of observation, measurement, and analysis appropriate for the data and tasks at hand, and on occasion, questions of suitable theoretical frameworks or of social context (Chrisman 1999). The corresponding forms of not knowing are considered in principle correctable because they assume that there is some true knowledge out there to be discovered given a chance for new observations, better methods, better theory, a more experienced analyst, less restrictive information sharing policies, a more open discussion of formerly taboo topics, and so on. Thus far work on geospatial uncertainty has focused overwhelmingly on these forms of not knowing stemming from diverse empirical shortcomings, and researchers have proposed a wide variety of ways to correct, mitigate or manage the many imperfections of geographic information products derived through common data analysis and modeling approaches. An international conference series, the 'Accuracy' symposia series, is devoted to these issues, and a steadily growing literature on the topic attests to its continued interest (Lowell and Jaton 1999, Hunsaker et al. 2001).

2.2 *That Which Cannot Be Known*

But what about that which *cannot* be known? What about uncertainty that is not a matter of incomplete, distorted, misunderstood, or mishandled information? This section focuses on that last and most problematic form of not knowing. Shifting the focus from data quality, or even product quality, to the quality of the knowledge produced with the help of GIS and related technologies will allow us to consider uncertainty in the geospatial domain within the broader context of the limits to knowledge in general.

Instances of intrinsic and irreducible limits to knowledge are well known from the formal sciences. Mathematicians have to live with the anomalies caused by the infinite and the infinitesimal and with a number of paradoxes of set theory studied by the mathematical philosopher Russell and others (Quine 1966). Their formal proofs are also subject to the limitations expressed in Gödel's incompleteness theorem. Gödel's theorem states that within any deductive system complex enough to contain elementary number theory, there will be some statements that cannot be proved to be either true or false. (These statements can be proved to be true or false within an expanded deductive system, but that system will have its own unprovable statements, and so *ad infinitum*). Gödel's theorem looms large in theoretical computer science, where uncomputability and undecidability are among the most vexing issues (Lewis and Papadimitriou 1981). Certain uncomputable problems, such as the famous 'tiling problem', are intrinsically spatial. The problem is as follows: given a finite set of regular, patterned tiles, and a set of rules regarding how tile patterns may be joined together, is it possible to determine whether that set of tiles can cover the plane without breaking any of the pattern-joining rules? The answer can be very easy to determine by trial and error in specific simple cases. However, the problem of defining a general procedure (algorithm) for determining whether any given finite set of tiles obeying a given set of rules will tile the plane is proved to be impossible to solve. This gets even stranger: The one-dimensional version of the problem, whereby an arbitrarily long and meandering 'domino snake' of tiles must connect two locations on the plane, is decidable on the infinite plane, but undecidable on the half-plane! (Harel 1987).

Physicists, for their part, have resigned themselves to the limits to knowledge imposed by the conundrums of quantum mechanics, Heisenberg's uncertainty principle, the breakdown of the laws of physics at space-time singularities and at the beginning of the Big Bang, and related paradoxes and anomalies. In the case of quantum mechanics a small minority of physicists believes that the problems must be due to 'hidden variables' and yet-to-be-discovered deeper levels of explanation but the mainstream Copenhagen interpretation holds that these uncertainties are in the nature of things. More disturbing to the ordinary scientist than the mysteries of the immensely large and the infinitesimally small should be the three-body problem in classical mechanics. It has been known since Newton's days that there is no analytic solution to the gravitational equations describing the dynamics of n bodies, where $n > 2$. Recent work has shown that the trajectories of the planets in our solar system are chaotic in the very long run, the mechanical perfection of the classical equations of motion being just a short-term (relatively speaking) approximation (Peterson 1990). Indeed, the mathematics of chaos and complexity theory have helped clarify what can and cannot be known in the case of complex nonlinear systems. The phenomenon of deterministic chaos in particular appears to offer one more reason why the future is generally *in principle* unknowable, regardless of our empirical ignorance of boundary conditions and the like. Wolfram (2002) and others have shown that deterministic chaos is a common behavior in systems modeled as cellular automata, a formalism now widely used by environmental scientists. Limits-of-knowledge problems are of course not restricted to the physical sciences, they are just more visible there because of the prevalence of formal approaches and proofs. Economics, the only formal social science, has its own impossibility theorems, of which Arrow's 'voting paradox' is the most famous. The theorem states that, given a small number of very reasonable requirements, it is not possible to find a function that will order options in a society in terms of the preferences of

its individual members (Arrow 1951). Developers of spatial decision support systems take heed!

Several of these areas of intrinsic ignorance in the formal sciences have their roots in a series of logical paradoxes, many of which had been known since Greek antiquity. Zeno's paradox, sometimes known as the 'Achilles and the Tortoise' problem, deals with the puzzle of infinity: If, in order to catch up with the tortoise, Achilles must first cover one-half the distance, then one-half of one-half, and so on, he'll take infinitely long to catch up! The paradox was resolved centuries later when it was discovered that the sum of an infinite series of intervals can be finite, but several other, more specialized puzzles involving infinity remain. Still very much with us and seemingly unresolvable is the famous 'Liar' paradox of Epimenides the Cretan who claimed that all Cretans were liars. If he spoke the truth, he must have been lying, and so on. The paradox illustrates the problem of self-reference, (in particular, self-contradiction) which also lies at the heart of Gödel's incompleteness theorem and other less well known issues with recursive functions. The Liar paradox is of potential relevance to knowledge production in the geospatial domain, as will be argued in the following section.

Another ancient puzzle known as the *sorites* paradox (from the Greek *soros*, meaning 'heap') has appeared in the GIScience literature a number of times in connection with vague representations (Bennett 2001). If 1 grain does not make a heap, 1+1 grains do not make a heap, 2+1 grains do not make a heap . . . then 9,999+1 grains do not make a heap either. The problem here is in the attempt to bridge the discrete concept of the individual grains with the continuous concept of a heap. Replace grains and heap with trees and forest, or buildings and town, and the logical diversion has something to say for the geospatial scientist. Formal semantics, drawing upon insights from linguistics and philosophy, has shown that vagueness is the inescapable punishment for cutting up a mostly continuous, heterogeneous and dynamic world into discrete, homogeneous and static categories (Williamson 1994, Haack 1996). Whether one uses fuzzy sets, rough sets, a multivalent logic, or some other device to tame the vagueness beast, no amount of additional or better data can resolve the uncertainty inherent in looking for boundaries (spatial as well as conceptual) where there are none.

As a final example, the notion of information itself, so central to the geographic information enterprise, has also lost some of the certainty imparted to it by Shannon's formalization. Deeper explorations of information theory have shown the qualitative information content of a message to be a relative, contextual quantity, contingent upon the prior knowledge of the recipient (Dretske 1981, Barwise and Seligman 1997). This implies that it is not possible to properly quantify the information content of a message if the recipient (or his/her state of knowledge) is not known, a situation that is clearly the rule rather than the exception.

Some of the issues discussed in this section have long been recognized in the GIScience literature and researchers have even sought to develop an ontology of 'imperfections' in geographic information (Duckham et al. 2001). Others may argue that, with few exceptions such as the problem of vagueness, these breaches of certainty are isolated curiosities in arcane corners of the formal sciences with little direct relevance to applied science. Yet the field of knowledge representation in artificial intelligence has come up with its own 'impossibility theorems' that stem directly from these formal limits to knowledge, while applying very generally to the everyday knowledge construction that geospatial researchers and practitioners engage in. These problems will be briefly reviewed in the following section.

3 Modes of Geospatial Knowledge Production

3.1 *Data, Method or Product: Which Comes First?*

This section argues that the prevailing modes of knowledge production in the geospatial domain are instances of certain general modes of commonsense reasoning and knowledge representation (KR) that have been formally studied in artificial intelligence (AI). Abduction and the many forms of non-monotonic reasoning are the most salient of these but deductive reasoning in empirical contexts has also been examined in the KR literature. The purpose of these studies is to understand the nature and limitations of knowledge-based systems. It appears that all these modes have intrinsic imperfections deriving from the formal limits to knowledge outlined in the previous section. None of these can be remedied with more or better data. In this section I discuss some apparent connections between these formal KR results and the kinds of practical knowledge construction that geospatial experts engage in.

In a paper on GIS objects with ill-defined boundaries, Couclelis (1996) distinguishes several hundred different cases where uncertain boundaries may crop up in an application, and these uncertainties may or may not practically matter in eight different ways. These estimates are derived from a framework that considers: (1) the empirical nature of the measurable characteristics of an object, (2) the mode of observation, measurement, or representation, and (3) the purpose for which the representation will be used. Ten different cases are distinguished along the first dimension, depending on whether the object considered is a thing or an extended entity, homogeneous or inhomogeneous, discontinuous or continuous, connected or distributed, solid or fluid, two- or three-dimensional, present or past, permanent or variable, fixed or moving, conventional or self-defining. These combine with six different aspects of observation and measurement (summarized as scale, resolution, projection, time, error, and theory) to yield a minimum of 120 different kinds of uncertainty in object boundaries. Very few of these are the result of error in the ordinary sense. Further, these uncertainties may be problematic, or not, depending on eight possible kinds of uses of the GIS object representations (exemplified by the user categories: managers of designed objects, agricultural land value assessors, fighter plane pilots, planners and resource managers, users of man-made facilities, graphic artists, bus and subway route map users, and space-time modelers). The main contribution of that paper is to shift the emphasis from an analysis of the ill-defined boundaries themselves to an examination of the conditions under which these emerge during the process of knowledge construction, and to their relative implications for the quality of knowledge obtained.

The knowledge construction perspective is even more clearly evident in Hootsmans (1996, p. 4) who describes three distinct approaches based on permutations of the three dimensions of data quality, method quality, and product quality. These dimensions closely parallel those in Couclelis (1996) briefly discussed above but go beyond the consideration of individual objects and their boundaries. These three approaches are summarized as follows:

- (1) In the *data-driven* approach, the data characteristics determine which processing method can be applied and what final product quality can be obtained. The processing of remotely sensed data is given as an example.
- (2) The *method-driven* approach starts with some standardized model of data processing, which imposes specific minimum requirements on the input data and determines

the quality of the product to the extent to which the data conform to the method's requirements. Examples are the application of a standardized land classification method or of a land suitability method.

- (3) In the *product-driven* approach the desirable final product is identified first and this determines the selection of appropriate methods which in turn impose their own requirements on the input data. Hootsmans cites as an example the production of topographic maps at a given scale but in fact most *process modeling* using pre-existing models appears to fit into that category.

The three approaches may be represented schematically as:

- (1) Data $\rightarrow$ method $\rightarrow$ product
- (2) Method $\rightarrow$ data $\rightarrow$ product
- (3) Product $\rightarrow$ method $\rightarrow$ data

In the following I argue that these schemes correspond to certain standard forms of commonsense reasoning studied in artificial intelligence. The first one represents *abduction*, the second, *deduction* in an empirical context, and the third encompasses diverse forms of *nonmonotonic reasoning*. Most reasonably complex research plans will involve not just one, but a combination of two or more of these schemes at different stages of the project. This will only compound the uncertainties, discussed below, inherent in each one of these approaches.

3.2 Commonsense Reasoning and Its Problems

In AI the term 'commonsense reasoning' encompasses practically all forms of reasoning except deduction in a formal context. This includes the various systematic modes of reasoning used in the empirical sciences and even formal deduction to the extent that it deals with uncertain premises, as is the case with probabilistic reasoning (Brachman et al. 1992). These modes revolve around a small number of notions that are well familiar to geospatial researchers and practitioners, such as: evidence (data, measurements, observations); methods, models and hypotheses; plausibility (probability, likelihood, odds, etc.); and outcomes (products, results, solutions, interpretations, diagnoses, explanations, predictions).

Abduction is a very common form of reasoning that covers model production and constraint satisfaction problems. Brachman et al. (1992) mention image interpretation right along with the classic example of medical diagnosis. Abductive reasoning produces a best explanation ('product') out of a set of observations ('data') such that the explanation satisfies a number of methodological or empirical constraints ('method'). In the geospatial domain the explanation may be an interpretation or model of a pattern seen in the data, which may be static, as in an image, or dynamic, as in the case of a meteorological or ecological process. A formal analysis of abduction is given in Bylander et al. (1992, p. 28), as follows:

"Abduction is a tuple $\langle D_{\text{all}}, H_{\text{all}}, e, pl \rangle$ where D_{all} is a finite set of all the data to be explained, H_{all} is a finite set of all the individual hypotheses, e is a map from subsets of H_{all} to subsets of D_{all} (H explains ($e(H)$)), and pl is a map from subsets of H_{all} to a partially ordered set (H has plausibility $pl(H)$)"

H is complete if it explains all the data. H is a *best* explanation if there is no explanation H' more plausible than H . The range of pl may be a classic or Bayesian

probability, a fuzzy value, a likelihood estimate, or any other measure with a partially ordered range. The authors prove that for composite explanations (made up of several hypotheses and sub-hypotheses) abduction is intractable (NP-hard) in general, and they identify a number of factors that lead to that result. In other words, the problem of finding the single *best* explanation or model for the data is intractable in all but trivial cases. (Some minor exceptions are discussed in that paper). What is more, additional information usually makes the problem even harder, unless it helps reduce the space of possible sub-hypotheses. Beven (2002a, b) comes to very similar conclusions in the context of hydrological modeling; namely, that there may be no way of uniquely identifying optimal model structures or parameters for a particular application, even when significant quantities of data are available for model calibration. Using reasoning closely related to the model of abduction described above, Beven proposes constructing a mapping (*e*) of a “landscape space” defined by the available empirical data (D_{all}) into a model space made up of a large number of model structures, along with various sets of parameter values and boundary conditions (H_{all}). These models may be viewed as multiple working hypotheses (*H*), some of which may be somewhat more plausible than others (*pl* (*H*)) on the basis of prior empirical evidence. Similar insights regarding the non-uniqueness of models derived from data are also obtained from formal theories of modeling and simulation (Zeigler et al. 2000).

At first blush the second, method-driven case appears much more secure. *Deduction* is the prototypically reliable reasoning mode and probability theory, from which all our statistical methods derive, is a formal deductive domain. Even though statistics cannot produce certain products with uncertain data, at least there is certainty in how data uncertainty may be quantified. Or is there? . . . In a classic paper McCarthy and Hayes (1969) distinguish between the *epistemological adequacy* of a system (its well-formedness with respect to truth-theoretic semantics) and its *heuristic adequacy* (the ability of its inferential procedures to arrive at desired conclusions). It turns out that the heuristic adequacy of method-driven products may be problematic to the extent that data quality information of different sorts, as well as data from different sources, are considered together. Here is what Grim et al. (1998, p. 53) have to say about this case that should be well familiar to geospatial researchers, who often need to take into account a variety of map quality and/or data quality information:

“Suppose that you receive information, of various shades of accuracy, from a number of sources with different track records of reliability. Some of the information you receive from these sources may be fairly straightforward. But some information you receive may itself be about the accuracy of other information or even about the general reliability of some of your other sources. This picture of conflicting information from conflicting sources . . . characterizes our epistemic lives quite generally. *Information sets within such a model can exhibit dynamic and chaotic phenomena quite similar to those introduced [in the previous pages].*” (Emphasis added.)

The authors ground that passage on their elaborations of the ‘Liar’ paradox mentioned in the previous section, involving systems of statements with partially conflicting mutual references. They show that such systems result in chaotic recursive loops that are strikingly illustrated by Grim et al. (1998) in their spatialised computational explorations of these phenomena.

Finally, in the product-driven mode, one starts with the choice of a previously defined formal or informal model (the required 'product'), which then determines the methods to be used and the needed type and quality of data. Even assuming that the chosen model is a 'good' one (a sometimes heroic assumption in the case of models of human or ecological processes), fitting the model to the world is an enterprise fraught with uncertainties of its own. Many of these go back to the difficulties of *nonmonotonic reasoning* which is the mode of reasoning best describing the product-driven mode of knowledge construction. Nonmonotonic reasoning involves the adoption of assumptions ('axioms') that may have to be abandoned in light of new information. Unlike classical deductive logic, nonmonotonic logics allow the set of theorems deriving from a set of axioms to decrease when new axioms are added to the system (Cohen and Feigenbaum 1982). An implemented process model derives a set of theorems (results, predictions) from a set of relations or rules ('axioms') reflecting how things are believed to work. The logic is nonmonotonic because adding new relations in a model is very likely to change the outcomes. This being the case, *a priori* process models are subject to the difficulties of all nonmonotonic reasoning, variously described in the AI literature under the terms inheritance hierarchies with exceptions, reasoning about prototypes, and **the frame problem in time-varying domains** (Brachman et al. 1992). At issue are questions such as: **What changes when a widely accepted general model is adapted to the conditions of specific places (and times)?** When do deductions from a good model, using good data, lead to bad results? Or: in a constantly changing world, which changes can be safely ignored by a particular dynamic model, and which cannot? These technical questions cannot be reviewed here but they do support the argument that model error can be a more substantial source of uncertainty than data error (Loague 2000).

How much do these epistemic subtleties matter in practice? Probably not an awful lot, or we would not have been able to even get started in the applied sciences. But the fact that they exist at all is a sobering reminder that uncertainty is an irreducible part of sufficiently complex knowledge. Bad data and human fallibility should not take all the blame.

4 Discussion and Conclusions

Uncertainty is built into the knowledge production process. Error in the geographic information domain is inevitable, and not just because of human limitations. This has implications for both the experts and for the public, including decision-makers. Within the expert community clearly the efforts to reduce and tame uncertainty in geospatial information must go on. The strong research program already in place, tackling uncertainty issues on so many different fronts, is likely to continue to yield substantial improvements in the quality and usability of geospatial information. But even for the most sophisticated and well-informed there will always be a residue of ignorance. It may thus help to put together a systematic 'encyclopedia of ignorance' for geospatial knowledge – to map out the ultimate *terra incognita* of epistemic impossibility – so that by recognizing the bounds of what cannot be known we may more effectively concentrate on the more tractable forms of not knowing. Logic, formal model theory and philosophy are likely to prove valuable sources of guidance and insights in such an endeavor. As Beven (2002a, b) and many others have shown, these abstract realms of understanding are very well compatible with strong empirical science.

Looking beyond the experts to the broader society, we can sympathize with Mowrer (1999, p. 3) who writes "Perhaps the worst nightmare of a natural resources manager is to appear 'uncertain' to the public, or to admit that there is 'error' in the decision process being presented." Still, I believe that the days when people thought that science meant certainty are well behind us. The nightmare today is not in the uncertainty itself but in its inadequate, unscientific expression: in the hand-waving, the wishy-washy statements, the ignorance concealed under pretty pictures, the pretense that something can be known that in fact cannot. There is too much of that going on. This is another reason why the ongoing program to pin down and communicate uncertainty affects much more than a small community of fastidious experts. These efforts are all the more important at an age when spatial data jump from the desktop to the handheld and ride unchecked along the web so that virtually anyone with the right technology can be an instant geospatial information provider or user. Goodness knows what new, unthought-of dimensions of uncertainty we may have to contend with under the currently developing geographic information use scenarios!

So, in closing: for all the undeniable progress, it seems that few things about geospatial knowledge are as certain as the truth encapsulated in C S Peirce's century-old aphorism: "Our knowledge always swims in a continuum of uncertainty and of indeterminacy" (cited in Menand 2001, p. 364).

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References

- Arrow K J 1951 *Social Choice and Individual Values* (Second edition). London, John Wiley and Sons
- Barwise J and Seligman J 1997 *Information Flow*. Cambridge, Cambridge University Press
- Bennett B 2001 Application of supervaluation semantics to vaguely defined spatial concepts. In Montello D R (ed) *Spatial Information Theory: Foundations of Geographic Information Science*. Berlin, Springer Lecture Notes in Computer Science No 2205: 108–23
- Beven K 2002a Towards an alternative blueprint for a physically based digitally simulated hydrologic response modelling system. *Hydrological Processes* 16: 189–206
- Beven K 2002b Towards a coherent philosophy for environmental modelling. *Proceedings of the Royal Society of London* 458: 2465–84
- Brachman R J, Levesque H J, and Reiter R 1992 *Knowledge Representation*. Cambridge, MA, MIT Press
- Bylander T, Allemang D, Tanner M C, and Josephson J R 1991 The computational complexity of abduction. In Brachman R J, Levesque H J, and Reiter R (eds) *Knowledge Representation*. Cambridge, MA, MIT Press: 25–60
- Chrisman N R 1999 Speaking truth to power: An agenda for change. In Lowell K and Jaton A (eds) *Spatial Accuracy Assessment: Land Information Uncertainty in Natural Resources*. Chelsea, MI, Ann Arbor Press: 27–31
- Cohen P R and Feigenbaum E A 1982 *The Handbook of Artificial Intelligence*. Los Altos, CA, William Kaufmann

- Couclelis H 1992 Geographic knowledge production through GIS: Towards a model for quality monitoring. In Couclelis H, Beard K, and Mackaness W (ed) *Two Perspectives on Data Quality*. Santa Barbara, CA, University of California, National Center for Geographic Information and Analysis Technical Report No 92-12: 1–39
- Couclelis H 1996 Towards an operational typology of geographic entities with ill-defined boundaries. In Burrough P A and Frank A U (eds) *Geographic Objects with Indeterminate Boundaries*. London, Taylor and Francis: 45–55
- Dretske F I 1981 *Knowledge and the Flow of Information*. Cambridge, MA, MIT Press
- Duckham M, Mason K, Stell J G, and Worboys M F 2001 A formal approach to imperfection in geographic information. *Computers, Environment and Urban Systems* 25: 89–103
- Grim P, Mar G, and St. Denis P 1998 *The Philosophical Computer: Exploratory Essays in Philosophical Computer Modeling*. Cambridge, MA, MIT Press/Bradford Books
- Haack S 1996 *Deviant Logic, Fuzzy Logic: Beyond the Formalism*. Chicago, University of Chicago Press
- Harel D 1987 *Algorithmics*. Wokingham: Addison-Wesley
- Hootsmans R M 1996 Fuzzy Sets and Series Analysis for Visual Decision Support in Spatial Data Exploration. Utrecht, University of Utrecht, PhD Dissertation
- Hunsaker C T, Goodchild M F, Friedl M, and Case T J 2001 *Spatial Uncertainty in Ecology: Implications for Remote Sensing and GIS Applications*. Berlin, Springer
- Lewis H R and Papadimitriou C H 1981 *Elements of the Theory of Computation*. Englewood Cliffs, NJ, Prentice-Hall
- Loague K 2000 Environmental modeling: The past, the present, and the future. In Heuvelink G B and Lemmens M J (eds) *Accuracy 2000: Proceedings of the Fourth International Symposium on Spatial Accuracy Assessment in Natural Resources and Environmental Science*. Delft, Delft University Press: 425–32
- Lowell K and Jaton A (eds) 1999 *Spatial Accuracy Assessment: Land Information Uncertainty in Natural Resources*. Chelsea, MI, Ann Arbor Press
- McCarthy J and Hayes P J 1969 Some philosophical problems from the standpoint of artificial intelligence. In Melzer B and Michie D (eds) *Machine Intelligence*. Edinburgh, Edinburgh University Press: 463–502
- Menand L 2001 *The Metaphysical Club: A History of Ideas in America*. New York, Farrar, Straus and Giroux
- Mowrer H T 1999 Accuracy (re)assurance: Selling uncertainty assessment to the uncertain. In Lowell K and Jaton A (eds) *Spatial Accuracy Assessment: Land Information Uncertainty in Natural Resources*. Chelsea, MI, Ann Arbor Press: 3–10
- Peterson I 1990 *Islands of Truth: A Mathematical Mystery Cruise*. New York, W H Freeman and Company
- Quine W V 1976 *The Ways of Paradox and Other Essays* (Revised and enlarged edition). Cambridge, MA, Harvard University Press
- Thrift N 1985 Of flies and germs: A geography of knowledge. In Gregory D and Urry J (eds) *Social Relations and Spatial Structure*. London, Macmillan: 366–403
- Williamson T 1994 *Vagueness*. London, Routledge
- Wolfram S 2002 *A New Kind of Science*. Champaign, IL, Wolfram Media
- Zeigler B P, Praehofer H, and Kim T G 2000 *Theory of Modeling and Simulation: Integrating Discrete Event and Continuous Complex Dynamic Systems* (Second edition). San Diego, CA, Academic Press